

HYDROGENATED NBR: PATENT LANDSCAPE ON HYDROGENATION, MOLECULAR-WEIGHT CONTROL, AND HNBR FORMULATION

1. ABSTRACT

This report synthesizes a set of patents relevant to hydrogenated nitrile butadiene rubber (HNBR), spanning upstream NBR polymerization, hydrogenation process control, catalyst selection/recovery, and downstream compounding. The target material is defined as nitrile rubber in which the butadiene-derived unsaturation is selectively reduced to produce partially or fully hydrogenated elastomeric products with controlled residual double-bond content, molecular weight, and viscosity. Across the selected patents, the dominant technical themes are: (1) solution hydrogenation of NBR using homogeneous noble-metal catalysts such as rhodium, ruthenium, or palladium systems; (2) simultaneous metathesis/hydrogenation concepts intended to reduce molecular weight while hydrogenating; (3) in-line or process analytical control to stop hydrogenation at a defined conversion; and (4) formulation strategies that exploit HNBR's thermal and chemical resistance in vulcanizates, conductive pastes, and adhesive systems.

The strongest synthesis-focused disclosures are EP2473281B1, which teaches simultaneous metathesis and hydrogenation of nitrile rubber in the presence of hydrogen using ruthenium/osmium complexes to obtain low-molecular-weight HNBR with narrower molecular-weight distribution, and US6307624B1, which discloses pressurized hydrogenation of NBR with homogeneous or heterogeneous catalysis while monitoring Raman spectra to determine the degree of hydrogenation in real time. EP3827029A1 addresses catalyst loading and post-hydrogenation purification, emphasizing ruthenium, palladium, or rhodium compounds at controlled ppm levels to improve ageing properties while minimizing residual noble metals. EP2292687B1 and US9321910B2 are formulation patents that define HNBR as a high-performance elastomer and specify hydrogenation degree, Mooney viscosity, and compounding systems for durable vulcanizates. US7795344B2 and US20180022951A1 are not direct HNBR synthesis patents, but they disclose hydrogenation of diene-containing nano-particles and the use of HNBR as a conductive-rubber component, respectively, showing how hydrogenated nitrile chemistry is integrated into advanced materials. Overall, the landscape favors solution-phase hydrogenation, noble-metal catalysis, careful control of residual unsaturation, and downstream stabilization/compounding to preserve ageing resistance and processability.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US8124234B2
- Patent Title: Polyurethane compositions and articles prepared therefrom, and methods for ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not directed to hydrogenated NBR synthesis. It discloses polyurethane-based compositions for adhesion and automotive interior applications, with elastomeric rubbers mentioned only as one possible component class in the broader composition space.

Relevant Experimental Evidence - Example a: discussed automotive sheet/skin performance requirements such as low gloss, heat resistance, and cold impact resistance, but no HNBR synthesis was demonstrated.

Technical Relevance This patent is only tangentially relevant because it references elastomer-rubber-containing compositions and automotive end uses, but it does not disclose HNBR preparation or hydrogenation chemistry.

Patent 2

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a process for producing low-molecular-weight hydrogenated nitrile rubber by carrying out nitrile-rubber metathesis and hydrogenation simultaneously in the presence of hydrogen. The process uses ruthenium- or osmium-based complexes and is intended to reduce molecular weight while hydrogenating the polymer, yielding HNBR with narrower molecular-weight distribution and lower residual double-bond content. - Catalyst amount: 0.041 phr — Example 1 (HNBR 1, Hoveyda

catalyst) - Catalyst amount: 0.045 phr — Example 1 (HNBR 2, Arlt catalyst) - Catalyst amount: 0.060 phr — Example 1 (HNBR 3, Wilkinson's catalyst) - Metal content: 65.6 ppm — Example 1 - Co-catalyst amount: 1.0 phr triphenylphosphine — Example 1 (HNBR 3) - Reaction time: 0 min, 60 min, 120 min, 180 min, 240 min — Example 1 hydrogenation profile - Residual double bond content: 2.5% — Example 1, HNBR 1 at 60 min - Residual double bond content: 0.6% — Example 1, HNBR 1 at 120 min - Residual double bond content: 0.4% — Example 1, HNBR 1 at 240 min - Residual double bond content: 7.0% — Example 1, HNBR 2 at 60 min - Residual double bond content: 4.5% — Example 1, HNBR 2 at 120 min - Residual double bond content: 3.5% — Example 1, HNBR 2 at 240 min - Residual double bond content: 15.4% — Example 1, HNBR 3 at 60 min - Residual double bond content: 3.1% — Example 1, HNBR 3 at 120 min - Residual double bond content: 0.7% — Example 1, HNBR 3 at 240 min - Mw: 113371 g/mol — Example 1, HNBR 1 after 240 min - Mn: 48352 g/mol — Example 1, HNBR 1 after 240 min - PDI: 2.34 — Example 1, HNBR 1 after 240 min - Mooney viscosity: 33.3 — Example 1, HNBR 1 after 240 min - Mw: 133758 g/mol — Example 1, HNBR 2 after 240 min - Mn: 54595 g/mol — Example 1, HNBR 2 after 240 min - PDI: 2.45 — Example 1, HNBR 2 after 240 min - Mooney viscosity: 41.3 — Example 1, HNBR 2 after 240 min - Mw: 195304 g/mol — Example 1, HNBR 3 after 240 min - Mn: 64130 g/mol — Example 1, HNBR 3 after 240 min - PDI: 3.05 — Example 1, HNBR 3 after 240 min - Mooney viscosity: 69.3 — Example 1, HNBR 3 after 240 min

Relevant Experimental Evidence - Example 1: compared three catalyst systems and showed that the disclosed catalyst gave faster hydrogenation and lower molecular weight than the industrial benchmark. - Example 1: demonstrated reduction of residual double bond content to 0.4% for HNBR 1 after 240 min. - Example 1: showed that the catalyst system can produce HNBR with lower Mooney viscosity and narrower PDI than the benchmark.

Technical Relevance This is a core HNBR synthesis patent because it directly addresses simultaneous hydrogenation and molecular-weight reduction of NBR. It is especially relevant for controlling residual unsaturation and viscosity in low-MW HNBR products.

Patent 3

- Patent Number: US9321910B2
- Patent Title: Microgel-containing vulcanizable composition based on hydrogenated nitrile ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not a synthesis patent for HNBR itself; it discloses vulcanizable compositions based on hydrogenated nitrile rubber, unsaturated carboxylic acids or salts, peroxide, and microgels. The HNBR is used as a high-temperature elastomeric base in compounded articles such as belts, hoses, gaskets, and

cables. - Mooney viscosity (ML 1+4 @ 100°C): 10 to 120 MU — Claim 1 - Degree of hydrogenation of diene units: 95 to 100% — Claim 1 - Unsaturated carboxylic acid/salt loading: 1 to 100 parts by weight per 100 parts rubber — Claim 1 - Peroxide loading: 0.2 to 8 parts by weight per 100 parts rubber — Claim 1 - Microgel glass transition temperature: less than or equal to -20°C — Claim 1 - Microgel loading: up to 60 parts by weight per 100 parts rubber — Claim 2 - Microgel glass transition temperature: less than or equal to -50°C — Claim 3 - Mooney viscosity (ML 1+4 @ 100°C): 15 to 100 MU — Claim 4 - Reaction temperature: 135°C — Example 1 (hydrogenation background discussion) - Hydrogenation pressure: 35 bar — Example 1 (background example) - Hydrogenation time: 6.5 h — Example 1 (background example)

Relevant Experimental Evidence - No numbered experimental examples were detected by the parser; the evidence block mainly contains formulation claims and background process discussion.

Technical Relevance Although not a synthesis disclosure, this patent is relevant because it defines the HNBR property window used in high-performance vulcanizates. It highlights the importance of high hydrogenation degree and controlled Mooney viscosity for downstream HNBR applications.

Patent 4

- Patent Number: EP2519350B1
- Patent Title: Highly active metathesis catalysis selective for romp and rcm
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses novel ruthenium transition-metal complexes and their use as selective metathesis catalysts for RCM, CM, ADMET, and ROMP. It also explicitly mentions use in depolymerization and hydrogenation-related processing of NBR/HNBR systems, including making depolymerized HNBR under hydrogen at elevated temperature and pressure. - Reaction temperature: 85°C — Example 1 - Ligand preparation scale: 5.0 mmol — Example 1 - Ligand preparation scale: 1.0 mmol — Example 1 - Catalyst precursor amount: 1.0 mmol — Example 1 (Ru complex synthesis) - CuCl amount: 2.5 mmol, 2.5 eq — Example 1 - Ligand amount: 1.2 mmol, 1.2 eq — Example 1 - Reaction time: 30-60 min — Example 1 and Example 2 (complex formation) - Ligand preparation yield: 86% — Example 1 - Ru complex yield: 4% — Example 1 - Ligand preparation yield: 91% — Example 2 - Ru complex yield: 29% — Example 2 - Ligand preparation yield: 54% — Example 3 - Ru complex yield: 5% — Example 3 - Ligand preparation yield: 62% — Example 4 - Ru complex yield: 32% — Example 4 - Ligand preparation yield: 63% — Example 6 - Ru complex yield: 14% — Example 6 - Ligand preparation yield: 56% — Example 7 - Ru complex yield: 8% — Example 7 - Ligand

preparation yield: 32% — Example 8 - Ru complex yield: 42% — Example 8 - Ligand preparation yield: 93% — Example 9 - Ru complex yield: 48% — Example 9 - Ligand preparation yield: 83% — Example 10 - Ru complex yield: 68% — Example 10 - Ligand preparation yield: 49% — Example 11 - Ru complex yield: 52% — Example 11 - Ligand preparation yield: 45% — Example 12 - Ru complex yield: 64% — Example 12 - Ligand preparation yield: 67% — Example 13 - Ru complex yield: 26% — Example 13 - Ligand preparation yield: 46% — Example 14 - Ru complex yield: 38% — Example 14 - Ligand preparation yield: 44% — Example 15 - Ru complex yield: 14% — Example 15 - Ligand preparation yield: 51% — Example 16 - Ru complex yield: 85% — Example 16 - Ligand preparation yield: 38% — Example 17 - Ru complex yield: 23% — Example 17 - Ligand preparation yield: 28% — Example 18 - Ru complex yield: 32% — Example 18 - Ligand preparation yield: 81% — Example 19 - Ru complex yield: 10% — Example 19 - Ru complex yield: 31% — Example 20 - Depolymerization temperature: 30-100°C — synthesis passage - Hydrogenation temperature: 60-150°C — synthesis passage - High-pressure hydrogen addition: high pressure — synthesis passage

Relevant Experimental Evidence - Example 1: prepared a Ru complex from a 5.0 mmol ligand precursor and obtained the target complex in 4% yield. - Example 2: showed an alternative Ru precursor route giving 29% yield of the Ru complex. - Example 7: demonstrated direct use of a ligand confirmed by LC-MS to make the Ru complex. - Example 20: showed a higher-yield Ru complex preparation route using an alternative Ru reagent. - Examples 1-20: collectively established a family of selective metathesis catalysts with tunable activity and selectivity.

Technical Relevance This patent is relevant because it provides catalyst chemistry that can be applied to NBR depolymerization and hydrogenation-related processing. While not a conventional HNBR hydrogenation patent, it supports molecular-weight reduction and hydrogen-assisted transformation routes for nitrile rubbers.

Patent 5

- Patent Number: US20180022951A1
- Patent Title: Electroconductive silver paste
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not a direct HNBR synthesis patent. It discloses electroconductive silver pastes in which the rubber phase can be NBR or HNBR, and it includes a preparation example where NBR is polymerized by emulsion polymerization and then hydrogenated to HNBR for use in stretchable conductive films. - Butadiene: 49 parts by weight — Preparation Example 1 - Acrylonitrile: 51 parts by weight — Preparation Example 1 - Deionized water: 270 parts by weight — Preparation Example 1 - Sodium

dodecylbenzenesulfonate: 0.5 part by weight — Preparation Example 1 - Condensed product of sodium naphthalenesulfonate: 2.5 parts by weight — Preparation Example 1 - tert-Dodecyl mercaptan: 0.3 part by weight — Preparation Example 1 - Triethanolamine: 0.2 part by weight — Preparation Example 1 - Sodium carbonate: 0.1 part by weight — Preparation Example 1 - Potassium persulfate: 0.3 part by weight — Preparation Example 1 - Hydroquinone: 0.5 part by weight — Preparation Example 1 - Polymerization temperature: 15°C — Preparation Example 1 - Polymerization time after initiator addition: 20 hours — Preparation Example 1 - NBR nitrile group content: 51% by weight — Example 1 / Example 3 - Alkali metal content: 50 ppm — Example 1 / Example 3 - A/B ratio: 81.5/18.5 — Example 1 / Example 3 - A/B ratio: 77.4/22.6 — Example 2 / Example 3 - Alkali metal content: 1400 ppm — Example 3 / Example 4 - Alkali metal content: 5500 ppm — Example 3 / Example 5 - NBR nitrile group content: 36% by weight — Example 3 / Example 4 - NBR nitrile group content: 26% by weight — Example 3 / Example 5 - HNBR nitrile group content: 50% by weight — Example 3 / Example 6 - HNBR alkali metal content: 55 ppm — Example 3 / Example 6 - HNBR iodine value: 25 mg/100 mg — Example 3 / Example 6

Relevant Experimental Evidence - Preparation Example 1: produced NBR by emulsion polymerization at 15°C using potassium persulfate and surfactant/mercaptan additives. - Preparation Example 1: converted the NBR to HNBR by hydrogenation (details not numerically specified in the excerpt). - Example 1: showed that an NBR with 51 wt% nitrile group and 50 ppm alkali metal gave very good conductivity and stretchability. - Example 3: compared NBR, HNBR, and other rubbers, showing that HNBR with 50 wt% nitrile group and 55 ppm alkali metal performed well. - Example 6: showed that HNBR-based conductive films retained good electrical and mechanical properties.

Technical Relevance This patent is relevant because it explicitly includes HNBR as a functional rubber phase and provides an upstream NBR polymerization example followed by hydrogenation. It also shows how HNBR properties such as nitrile content and alkali-metal content affect downstream performance in conductive formulations.

Patent 6

- Patent Number: EP2292687B1
- Patent Title: ... containing ethylene acrylate (AEM), polyacrylate (ACM) and/or hydrogenated ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is primarily a formulation patent for guanidine-free vulcanizable mixtures based on AEM, ACM, and/or HNBR rubbers. For HNBR, it discloses the rubber composition, hydrogenation level, and curing/retarder systems

rather than the hydrogenation synthesis itself. - Hydrogenated nitrile rubber loading: 100 parts by weight per 100 parts rubber — Claim 1 - Mooney viscosity (ML 1+4 @ 100°C): 10 to 120 MU — Claim 1 - Degree of hydrogenation of diene units: 95 to 100% — Claim 1 - Unsaturated carboxylic acid/salt loading: 1 to 100 parts by weight per 100 parts rubber — Claim 1 - Peroxide loading: 0.2 to 8 parts by weight per 100 parts rubber — Claim 1 - Microgel loading: up to 60 parts by weight per 100 parts rubber — Claim 2 - Mooney viscosity (ML 1+4 @ 100°C): 15 to 100 MU — Claim 4 - Hydrogenation temperature: 138°C — Example 2 - Hydrogenation pressure: 85 bar — Example 2 - Hydrogenation time: 4 hrs — Example 2 - NBR1 butadiene content: 60.4% — Table A, Example 2 - NBR1 maleic acid monoethyl ester content: 4.5% — Table A, Example 2 - NBR1 acrylonitrile content: 35.1% — Table A, Example 2 - NBR1 Mooney viscosity (ML 1+4 @100°C): 35 — Table A, Example 2 - HNBR1 residual double bond content: 0.4% — Example 2 - HNBR1 maleic monoethylester content: 4.5% — Example 2 - HNBR1 Mooney viscosity (ML 1+4 @100°C): 101 — Example 2

Relevant Experimental Evidence - Example 1: compared guanidine-containing and guanidine-free accelerator systems in ACM-based vulcanizates. - Example 2: described hydrogenation of NBR1 at 138°C and 85 bar for 4 h to obtain HNBR1. - Example 2: showed HNBR1 with 0.4% residual double bond content and Mooney viscosity of 101. - Example 2: demonstrated that retarder addition affects vulcanization kinetics in HNBR-containing compounds.

Technical Relevance This patent is relevant because it defines HNBR as a high-performance rubber and provides a concrete hydrogenation example with residual unsaturation and viscosity data. It also shows how HNBR is formulated for curing and high-temperature service.

Patent 7

- Patent Number: US6307624B1
- Patent Title: Process for the preparation of partially hydrogenated acrylonitrile-butadiene ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2001
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses partial hydrogenation of NBR in a pressurized reactor using homogeneous or heterogeneous catalysis, with Raman spectroscopy used in situ to monitor the degree of hydrogenation. The process is designed to stop the reaction at a target residual double-bond content, improving reproducibility and space-time yield. - NBR charge: 225 g — Example 1 - Acrylonitrile content: approx. 40% — Example 1 - Butadiene content: approx. 60% — Example 1 - 1,4-trans butadiene units: 85.5% — Example 1 - 1,4-cis butadiene units: 7.3% — Example 1 - 1,2-vinyl butadiene units: 7.2% — Example 1 - Triphenylphosphine: 2.25 g — Example 1 - Rhodium catalyst: 0.675 g — Example 1 - Chlorobenzene: 1275 g — Example 1 - Autoclave volume: 2 liter — Example 1 - Agitator speed:

280 min⁻¹ — Example 1 - Reaction temperature: 135°C — Example 1 - Hydrogen pressure: 35 bar — Example 1 - Raman scans: 200 scans — Example 1 - Spectral resolution: 4 cm⁻¹ — Example 1 - Hydrogenation time: 6.5 h — Example 1 - Residual double bond content: 19.2% — Example 1

Relevant Experimental Evidence - Example 1: demonstrated in situ Raman monitoring of NBR hydrogenation in a 2 L autoclave. - Example 1: showed hydrogenation at 135°C and 35 bar with a rhodium/phosphine catalyst system. - Example 1: calculated a residual double-bond content of 19.2% after 6.5 h, illustrating process control.

Technical Relevance This is a key HNBR synthesis patent because it directly addresses controlled hydrogenation of NBR to partially hydrogenated HNBR. Its main contribution is process analytics: real-time Raman monitoring to stop hydrogenation at the desired conversion.

Patent 8

- Patent Number: US7795344B2
- Patent Title: Nano-particle preparation and applications
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses polymer nano-particles based on poly(alkenylbenzene) cores with conjugated-diene or polyalkylene surface layers, and it explicitly states that the poly(conjugated diene) layer can be optionally hydrogenated. The hydrogenation is applied to the nanoparticle surface layer rather than to bulk NBR, but it demonstrates hydrogenation of diene-containing polymer domains in a controlled morphology. - Reactor jacket temperature: 26°C — Example 1 - Styrene charge: 1.36 kg (33 wt %) — Example 1 - Hexane charge: 0.90 kg — Example 1 - DVB charge: 50 mL — Example 1 - OOPs solution: 1.0 mL — Example 1 - Reactor jacket temperature: 50°C — Example 1 - Exotherm peak temperature: 54°C — Example 1 - Average particle diameter: about 50 nm — Example 1 - Particle polydispersity: about 1.1 — Example 1 - Butadiene/hexane blend: 1.04 kg at 21.6 wt% butadiene — Example 8 - Polymerization temperature: 57°C — Example 8 - Butyllithium solution: 5.4 mL of 1.5 M — Example 8 - Hydrogenation temperature: 120°C — Example 8 - Hydrogen pressure: about 792 kPa — Example 8 - Hydrogenation conversion: about 95% — Example 8 - Average particle size: ~30 nm — Example 8 - Isoprene/hexane blend: 1.5 kg at 14.8 wt% isoprene — Example 9 - Polymerization temperature: 58°C — Example 9 - Butyllithium solution: 5.4 mL of 1.5 M — Example 9 - Hydrogenation conversion: 92% — Example 10 - Average particle size: ~30 nm — Example 10 - Hydrogenation conversion: 69% — Example 19 - Hydrogenation conversion: 89% — Example 20

Relevant Experimental Evidence - Example 1: formed nano-particles with an average diameter of about 50 nm. - Example 8: hydrogenated the butadiene-containing particle phase at 120°C and about 792 kPa H₂ to about 95% conversion. - Example 9: prepared isoprene-based nano-particles at 57-58°C with controlled particle size. - Example 10: further hydrogenated the nano-particles to 92% conversion. - Example 20: showed that increased hydrogen charging raised hydrogenation conversion to about 89%.

Technical Relevance This patent is indirectly relevant because it demonstrates hydrogenation of conjugated-diene polymer domains in a controlled nano-particle architecture. It is useful as supporting evidence for selective hydrogenation chemistry, though not for conventional bulk HNBR manufacture.

Patent 9

- Patent Number: US11106161B2
- Patent Title: Intermediate transfer members for use with indirect printing systems and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not a hydrogenated NBR synthesis patent. It discloses curable polymer compositions and intermediate transfer members for printing, with cured acrylic rubber (ACM) substrates and silicone release layers; HNBR is not the central material system. - Adhesive layer thickness: 1 to 5 micrometer — Example 1 - ACM rubber sheet size: 20 cm by 20 cm — Example 1 - Curing temperature: 120°C — Example 1 - Curing time: 5 minutes — Example 1 - Fluid curable layer thickness: 5 to 100 µm — Example 1 - Room-temperature cure: 20 hours — Example 1 - Post-cure temperature: 140°C — Example 1 - Post-cure time: 1 hour — Example 1 - Adhesive composition 3 curing temperature: 90°C — Example 3 - Adhesive composition 3 curing time: 2 minutes — Example 3 - Adhesive composition 4 curing temperature: 100°C — Example 4 - Adhesive composition 4 post-cure sequence: 80°C for 1 h, 120°C for 1 h, 150°C for 1 h — Example 4 - Adhesion rating: 4 excellent — Example 1, Example 3, Example 12, Example 13, Example 14 - Adhesion rating: 3 good — Example 4, Example 6, Example 8, Example 9, Example 10, Example 11 - Adhesion rating: 1 poor — Example 5, Example 7

Relevant Experimental Evidence - Example 1: showed excellent adhesion between ACM rubber and a silicone elastomer using a thermally activated adhesive layer. - Example 3: demonstrated that peroxide-activated adhesive chemistry can also give excellent adhesion. - Example 4: showed a multi-step cure schedule at 80°C, 120°C, and 150°C for a silicone release layer. - Example 5: showed poor adhesion when water remained in the adhesive system. - Example 12: demonstrated excellent adhesion using a 3-aminopropyltriethoxysilane-based adhesive.

Technical Relevance This patent is only marginally relevant to HNBR because it focuses on ACM/silicone printing materials rather than nitrile rubber hydrogenation. It is included only because the broader search set captured elastomeric rubber processing and adhesive-layer chemistry.

Patent 10

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses hydrogenation of unsaturated nitrile rubber in solution using ruthenium, palladium, or rhodium compounds, with emphasis on using controlled catalyst amounts and avoiding difficult post-removal steps. It targets HNBR with improved ageing properties and reduced noble-metal residues, and it explicitly frames the process as a direct route to HNBR compositions. - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200°C — Claim 1 - Hydrogenation pressure: 700000 pascals to 15000000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - HNBR composition ageing: gel content increase less than 10% after 4 days at 140°C — Description - HNBR composition ageing: Mooney viscosity increase less than 40% after 4 days at 140°C — Description

Relevant Experimental Evidence - No numbered experimental examples were detected by the parser; the evidence block is dominated by claims and detailed process discussion.

Technical Relevance This is one of the most directly relevant patents for HNBR synthesis because it specifies catalyst loading, temperature, pressure, and time windows for solution hydrogenation. It also addresses a major industrial issue in HNBR manufacture: residual noble-metal removal and ageing stability.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2473281B1 and EP3827029A1 both target HNBR process optimization, but EP2473281B1 emphasizes simultaneous metathesis/hydrogenation for molecular-weight reduction, whereas EP3827029A1 emphasizes conventional solution hydrogenation with controlled noble-metal ppm levels and catalyst removal concerns.
- US6307624B1 focuses on partial hydrogenation control using in situ Raman spectroscopy, while EP3827029A1 focuses on catalyst loading and ageing performance; both address process reproducibility, but by different control strategies.
- EP2292687B1 and US9321910B2 are downstream formulation patents: both define HNBR as a high-temperature elastomer, but EP2292687B1 gives a concrete hydrogenation example (138°C, 85 bar, 4 h) whereas US9321910B2 mainly specifies composition windows and microgel/peroxide curing.
- US20180022951A1 includes an upstream NBR emulsion polymerization example and then hydrogenation to HNBR for conductive films; unlike the core synthesis patents, its HNBR use case is functional materials rather than bulk elastomer production.
- US7795344B2 demonstrates hydrogenation of diene-containing nano-particle surface layers at 120°C and about 792 kPa H₂, which is mechanistically related to selective hydrogenation but not to conventional NBR-to-HNBR manufacturing.
- Catalyst families differ across the synthesis patents: EP2473281B1 uses ruthenium/osmium metathesis complexes, US6307624B1 uses rhodium/phosphine or heterogeneous catalysts, and EP3827029A1 allows ruthenium, palladium, or rhodium compounds.
- Process temperatures span from 60-200°C in EP3827029A1, 135°C in US6307624B1, and 138°C in EP2292687B1, indicating that industrial HNBR hydrogenation commonly operates in the 100-150°C range.
- Residual unsaturation control is explicit in EP2473281B1 (RDB values down to 0.4%) and US6307624B1 (Raman-based endpoint control), while EP3827029A1 emphasizes ageing stability and low residual catalyst rather than direct RDB monitoring.

4. CONCLUSION

Across the patent set, the most actionable HNBR synthesis guidance comes from solution-phase hydrogenation patents that tightly control catalyst type, catalyst loading, temperature, pressure, and endpoint. EP3827029A1 provides the broadest industrial operating window for direct HNBR hydrogenation with ruthenium/palladium/rhodium catalysts, while US6307624B1 shows that in situ Raman monitoring can improve reproducibility of partial hydrogenation. EP2473281B1 adds a distinct route in which metathesis and hydrogenation are combined to lower molecular weight and narrow dispersity, which is valuable when lower-viscosity HNBR is desired. The evidence supports operating in the 60-150°C range, under elevated hydrogen pressure, with catalyst loadings in the low-ppm to low-phr regime depending on catalyst class, and with explicit control of residual double bonds and post-reaction catalyst removal. For downstream performance, the patents consistently indicate that high hydrogenation degree, controlled Mooney viscosity, and low residual metal content are important for ageing resistance and processability.

5. REFERENCES

- US8124234B2 | Polyurethane compositions and articles prepared therefrom, and methods for ... | Not disclosed | US | 2012 | <https://patents.google.com/patent/US8124234B2/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US9321910B2 | Microgel-containing vulcanizable composition based on hydrogenated nitrile ... | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9321910B2/en>
- EP2519350B1 | Highly active metathesis catalysis selective for romp and rcm | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2519350B1/en>
- US20180022951A1 | Electroconductive silver paste | Not disclosed | US | 2018 | <https://patents.google.com/patent/US20180022951A1/en>
- EP2292687B1 | ... containing ethylene acrylate (AEM), polyacrylate (ACM) and/or hydrogenated ... | Not disclosed | EP | 2012 | <https://patents.google.com/patent/EP2292687B1/en>
- US6307624B1 | Process for the preparation of partially hydrogenated acrylonitrile-butadiene ... | Not disclosed | US | 2001 | <https://patents.google.com/patent/US6307624B1/en>
- US7795344B2 | Nano-particle preparation and applications | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7795344B2/en>
- US11106161B2 | Intermediate transfer members for use with indirect printing systems and ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11106161B2/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>